



CS6140: Machine Learning

Neural Networks

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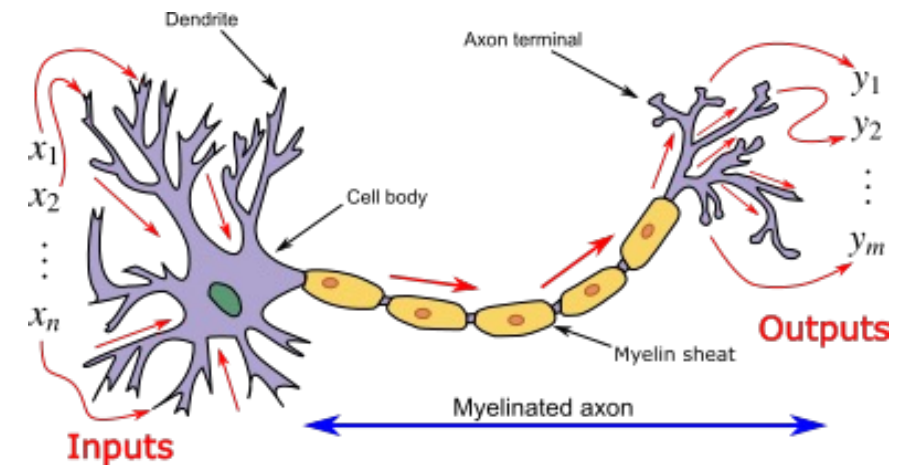
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Northeastern
University

What is a Neuron?

- A biological neuron is the basic unit of the human brain, which is specially designed to transmit information to other nerve cells, muscle, etc. This is done by a network consisting of billions of connected neurons.
- A neuron takes some **inputs** (through dendrites), processes them, and provides **outputs** (through axons).
- If the combination of input electrical signals is above a certain threshold, it results in **firing or activation of the neuron**.
- The same principle of firing/activation is used in artificial neurons as well. The Artificial Neural Networks (**ANN**) is inspired by this arrangement of biological neurons in our brains and its basic unit is called a neuron or a *Perceptron*.



Neural Networks

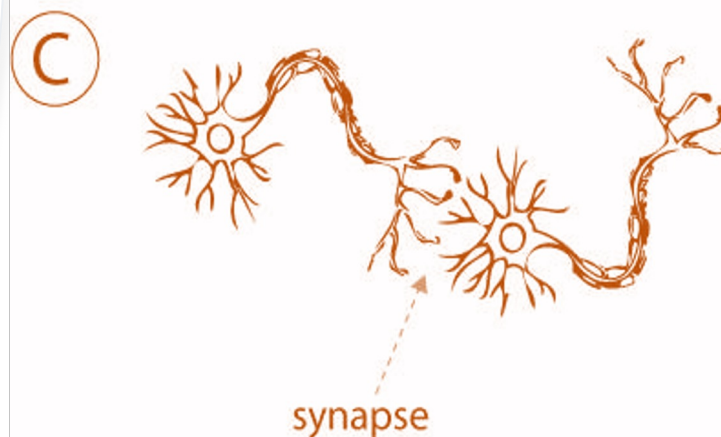
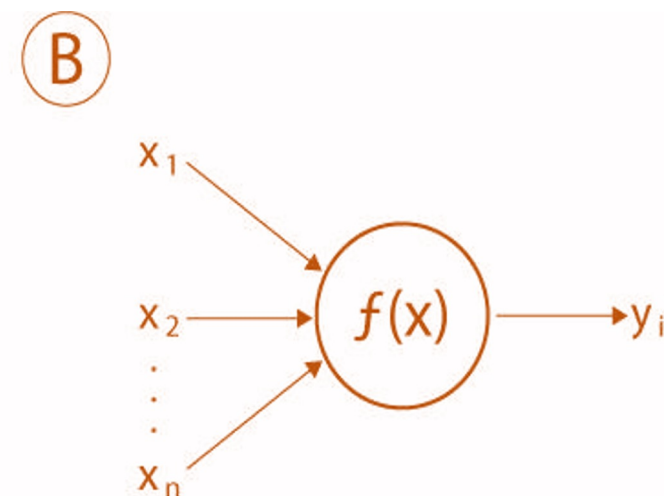
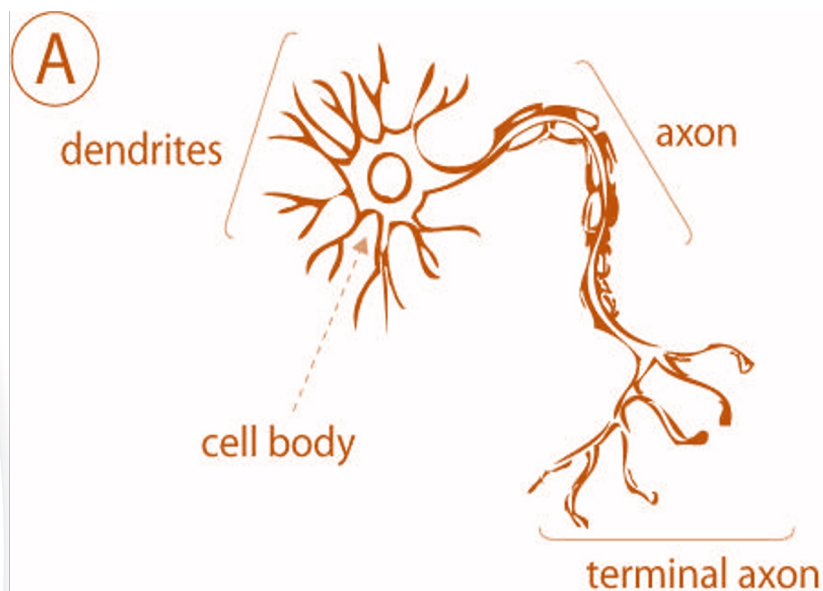
“Neural network - a beautiful biologically-inspired programming paradigm which enables a computer to learn from observational data.”

Michael Nielsen

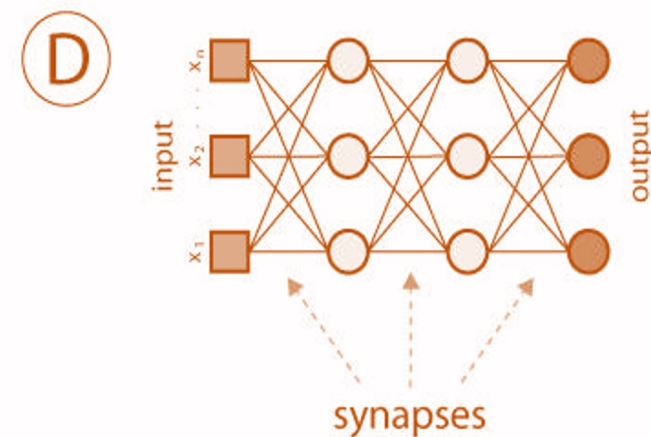
A short branched extension of a nerve cell, along which impulses received from other cells.



Activation function receives the outputs of previous layer

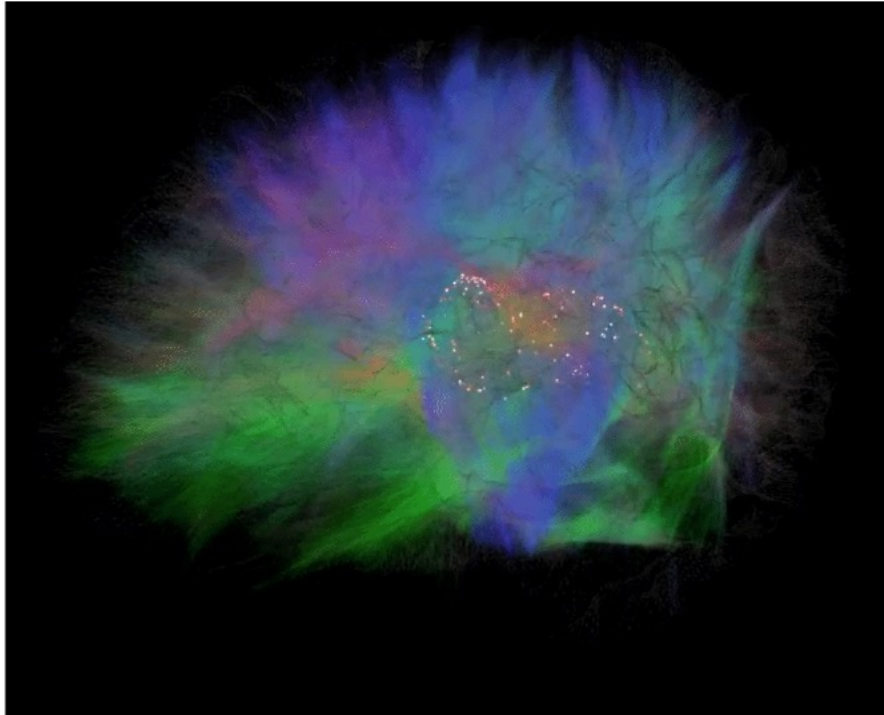


A junction between two nerve cells



Connection between neurons in two adjacent layers

Human Brain vs ANN



Human Brain

- **Thalamocortical system:**
3 million neurons
476 million synapses
- **Full brain:**
100 billion neurons
1,000 trillion synapses

Artificial Neural Network

- **ResNet-152:**
60 million synapses

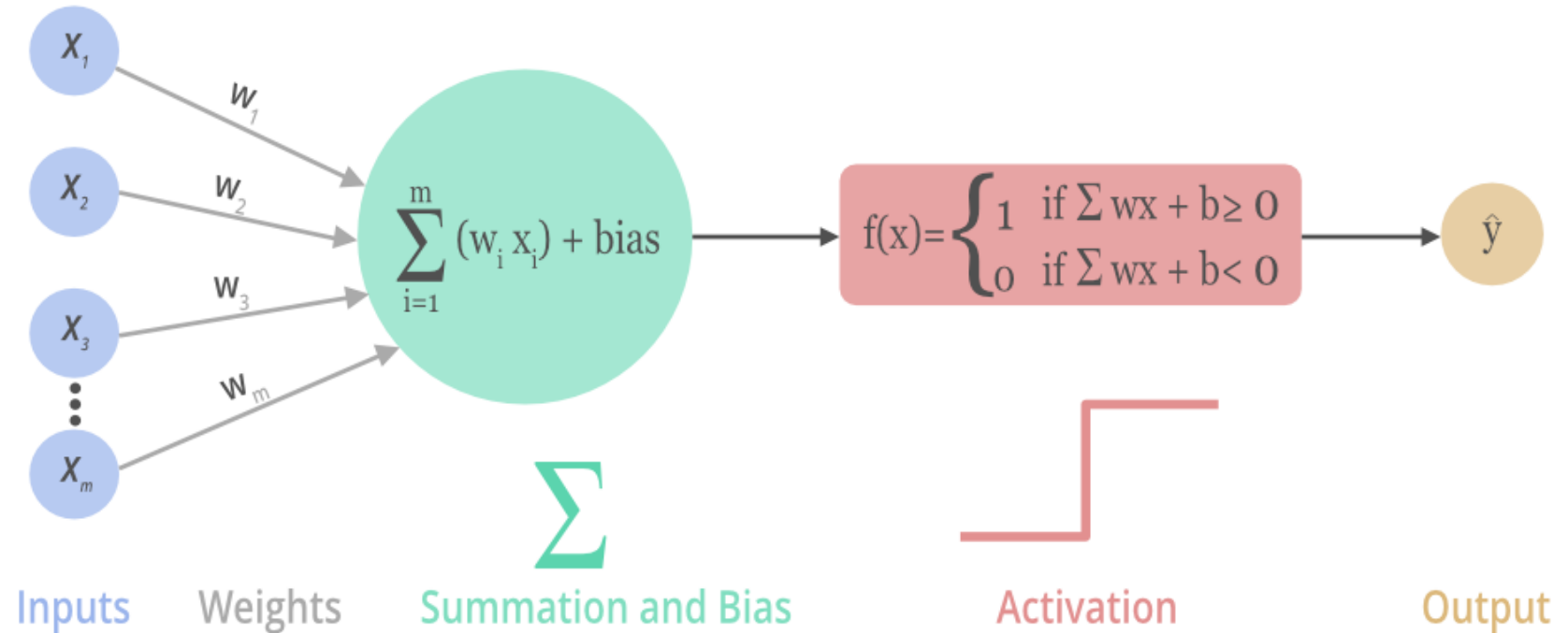
Human brains have ~10,000,000 times synapses than artificial neural networks.

By Lex Fridman: <https://deeplearning.mit.edu/>

Main components of ANN

- Receives input from n sources
- Computes weighted sum $h_1 = x_1w_1 + x_2w_2 + \dots x_nw_n$
- Passes through an activation function

The FIVE
Components:



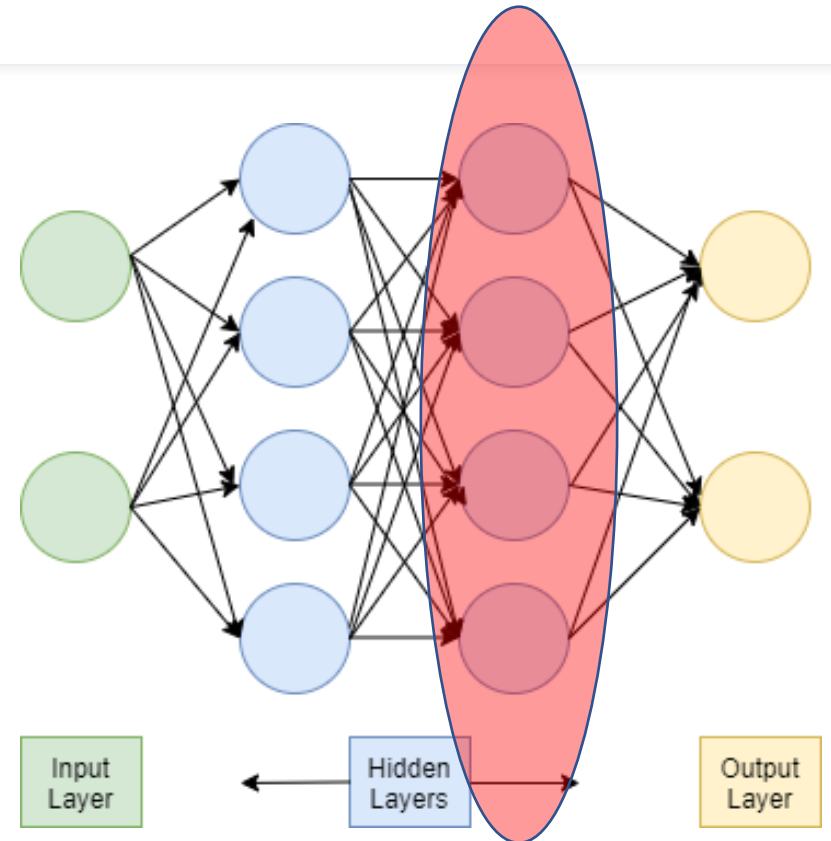
Types of Layers in ANN

There are 3 types of layers:

1. Input Layer: This layer is used to feed the input, eg:- if your input consists of 2 numbers, your input layer would have 2 nodes.

2. Output Layer: This is the output layer of the network. If a classification model's job is to predict between 5 classes, this layer will contain 5 nodes.

3. Hidden Layer(s): The hidden layers are the layers between the input and output layers. If there are two or more hidden layers, it is called as a **Deep Neural Network**.



Types of Layers in ANN

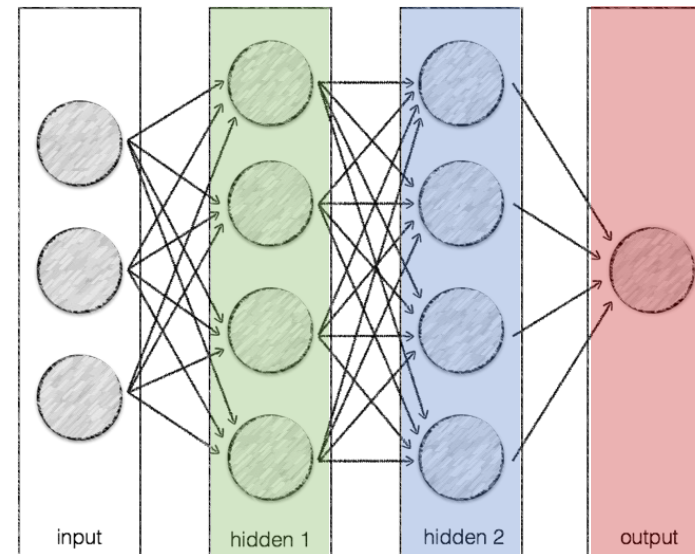
Composition of Functions
 $(f \circ g)(x) = f(g(x))$

Input = input

Hidden layer 1 = f (input)

Hidden layer 2 = f (f (input)) or f (Hidden layer 1)

Output = f (f (f (input))) or f (Hidden layer 2)

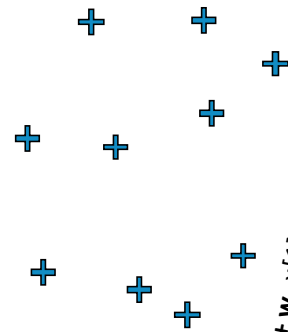


Perceptron

Remember the linear classifier based on score

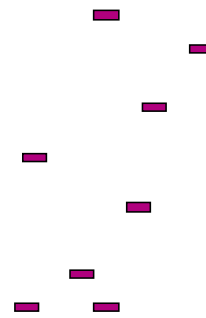
$$\text{Score}(x) = w_0 + w_1 x[1] + w_2 x[2] + \dots + w_d x[d]$$

$\text{Score}(x) > 0$



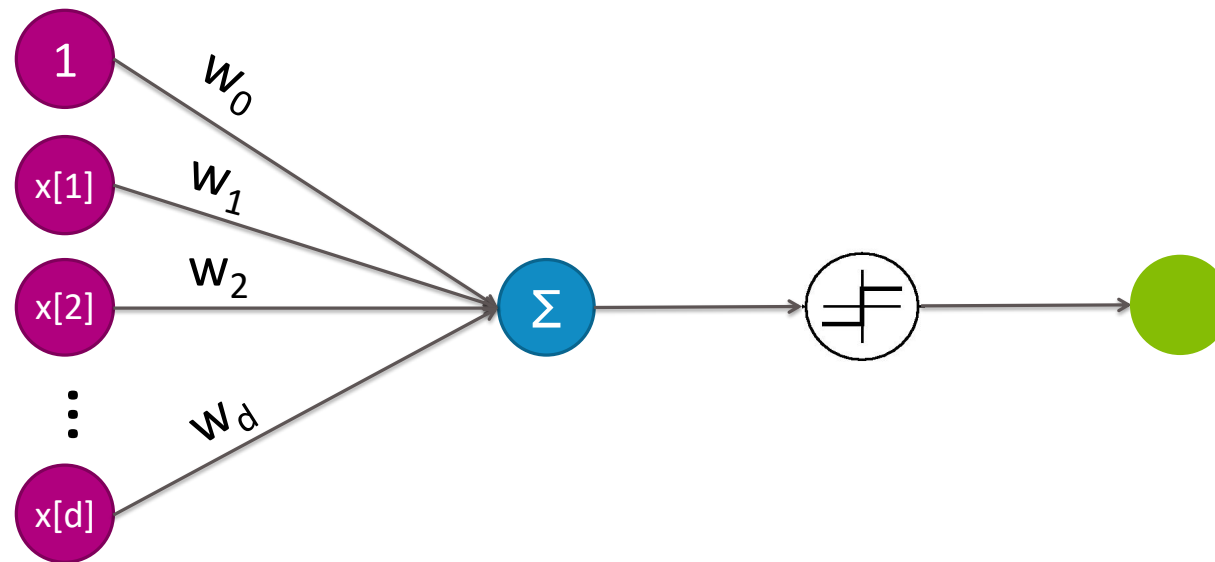
$$w_0 + w_1 x[1] + w_2 x[2] + \dots + w_d x[d] = 0$$

$\text{Score}(x) < 0$



Perceptron

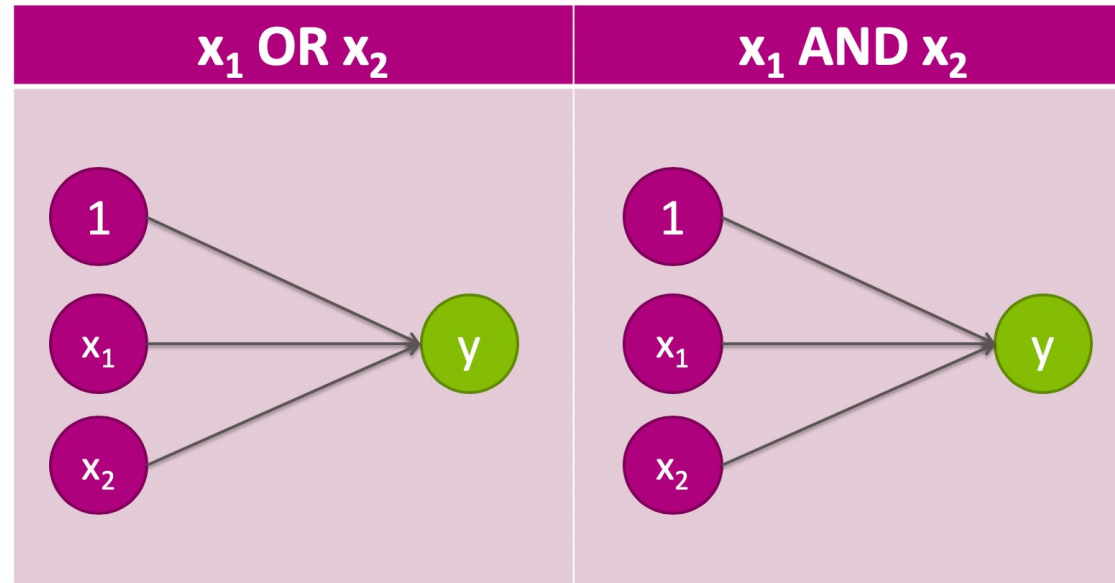
Graphical representation of this same classifier



This is called a **perceptron**

Perceptron

$w_0 =$
 $w_1 =$
 $w_2 =$



x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	1

x_1	x_2	y
0	0	0
0	1	0
1	0	0
1	1	1

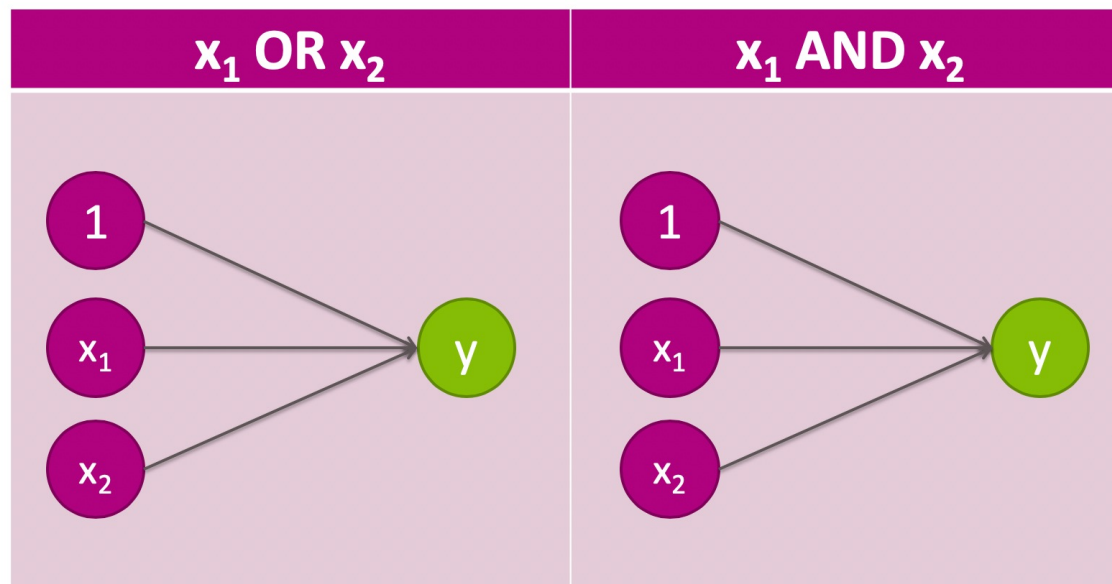
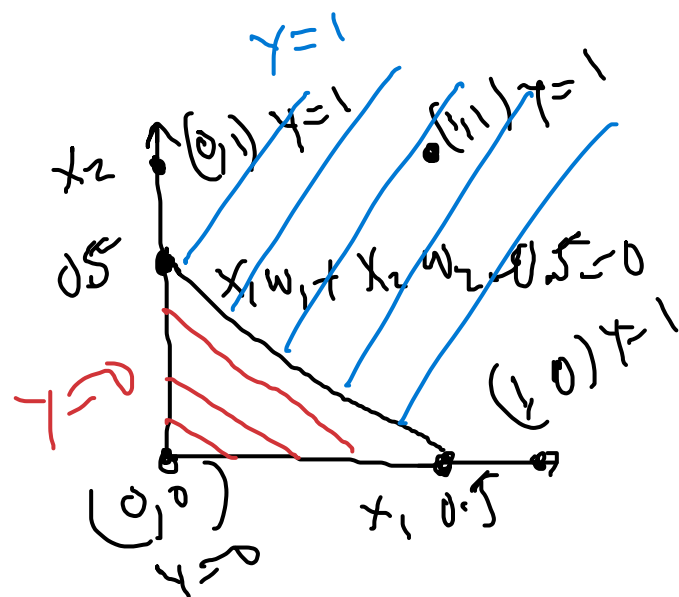
$w_0 =$
 $w_1 =$
 $w_2 =$

Perceptron

$$w_0 = -0.5$$

$$w_1 = 1$$

$$w_2 = 1$$



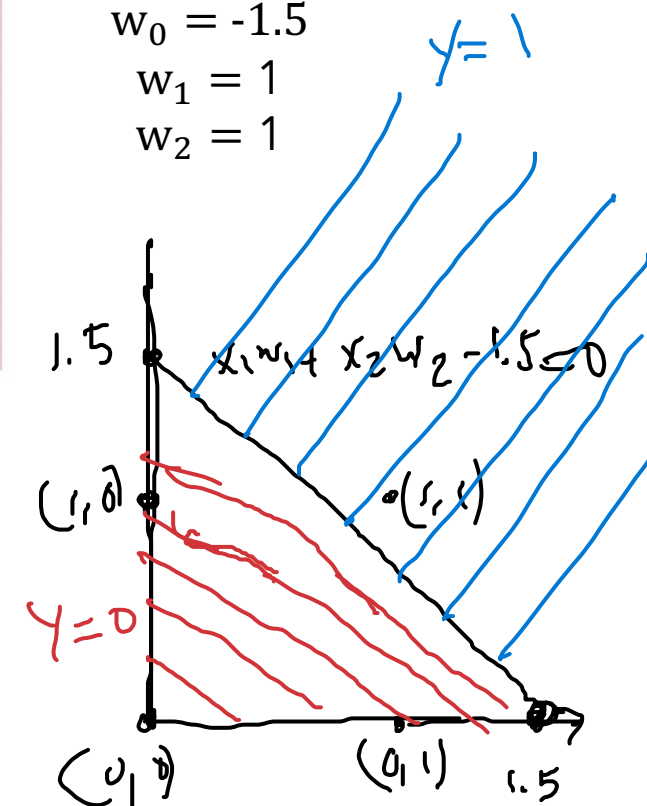
x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	1

x_1	x_2	y
0	0	0
0	1	0
1	0	0
1	1	1

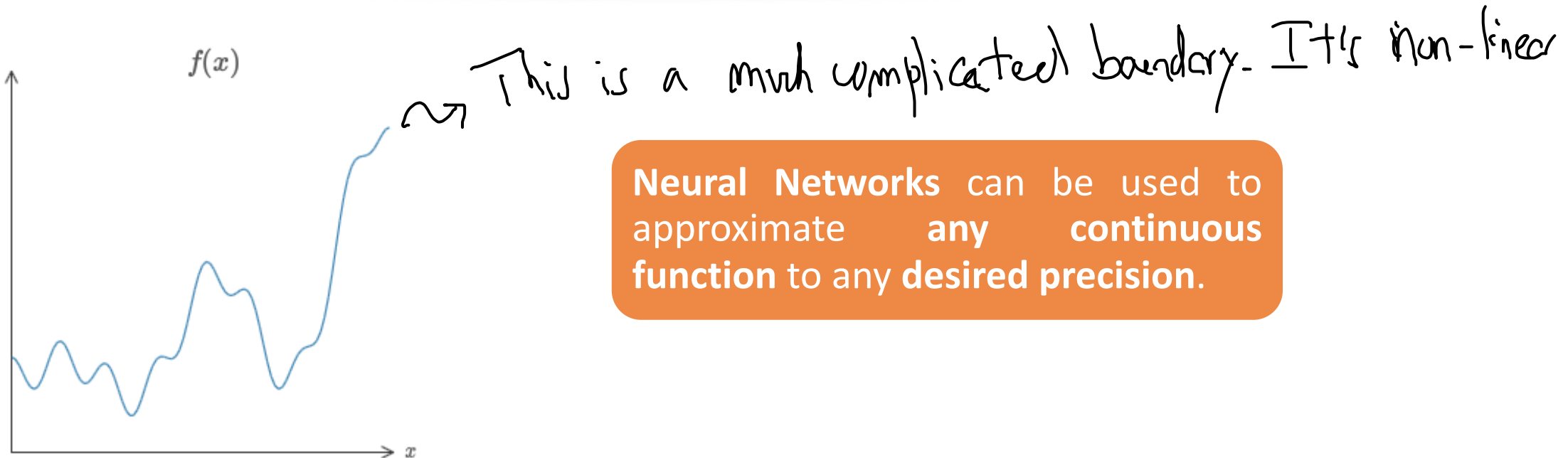
$$w_0 = -1.5$$

$$w_1 = 1$$

$$w_2 = 1$$



Universal Approximation Theorems

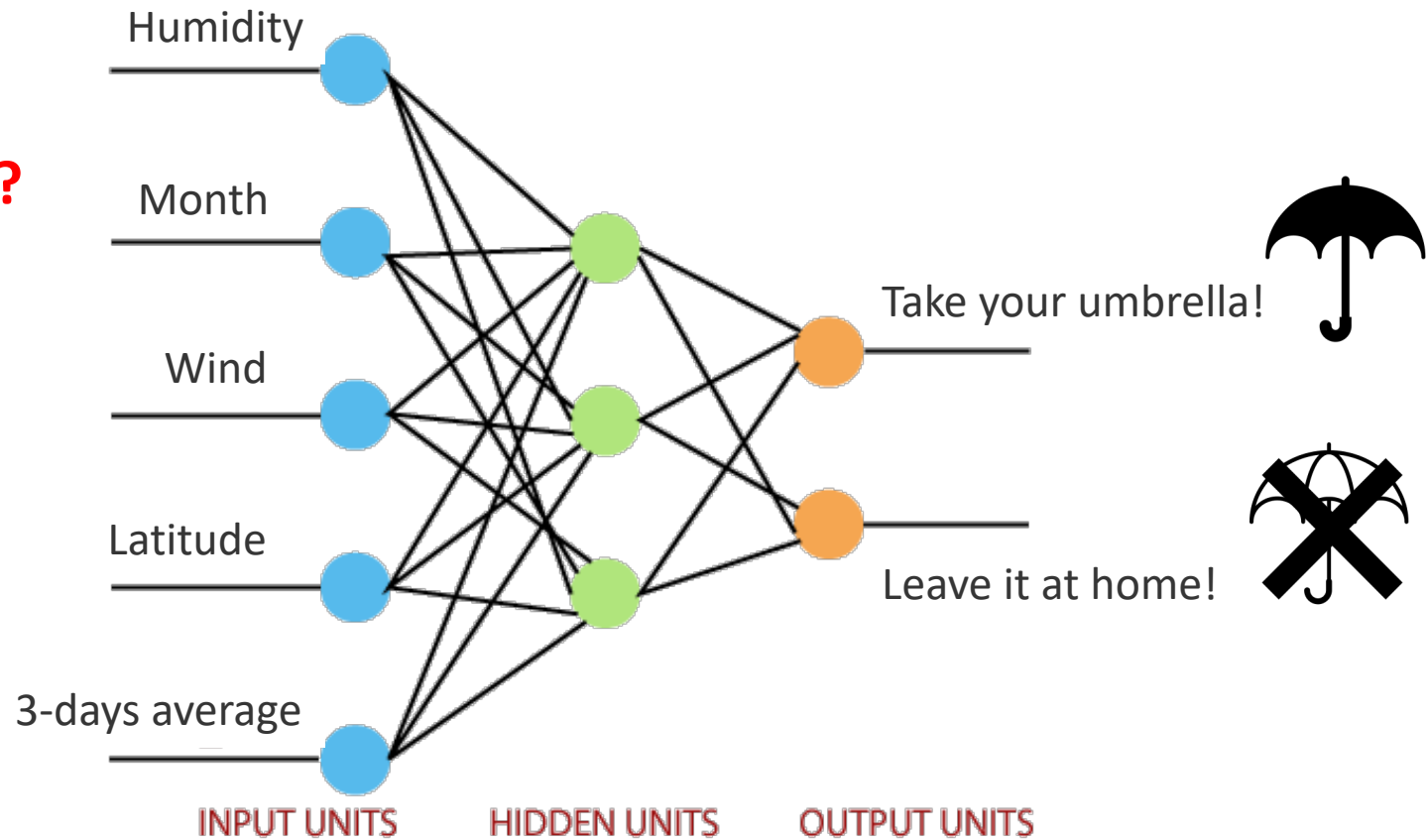
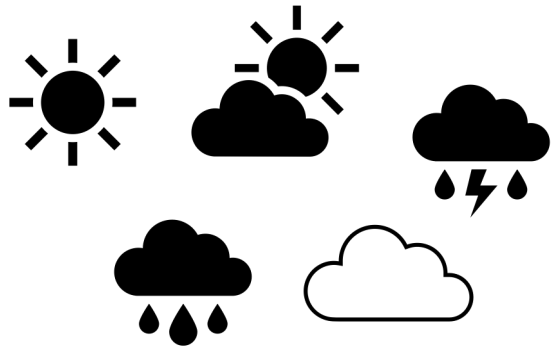


Neural Networks can be used to approximate **any continuous function** to any desired precision.

Visual Proof: <http://neuralnetworksanddeeplearning.com/chap4.html>

Example (weather forecast)

Do I need umbrella tomorrow?



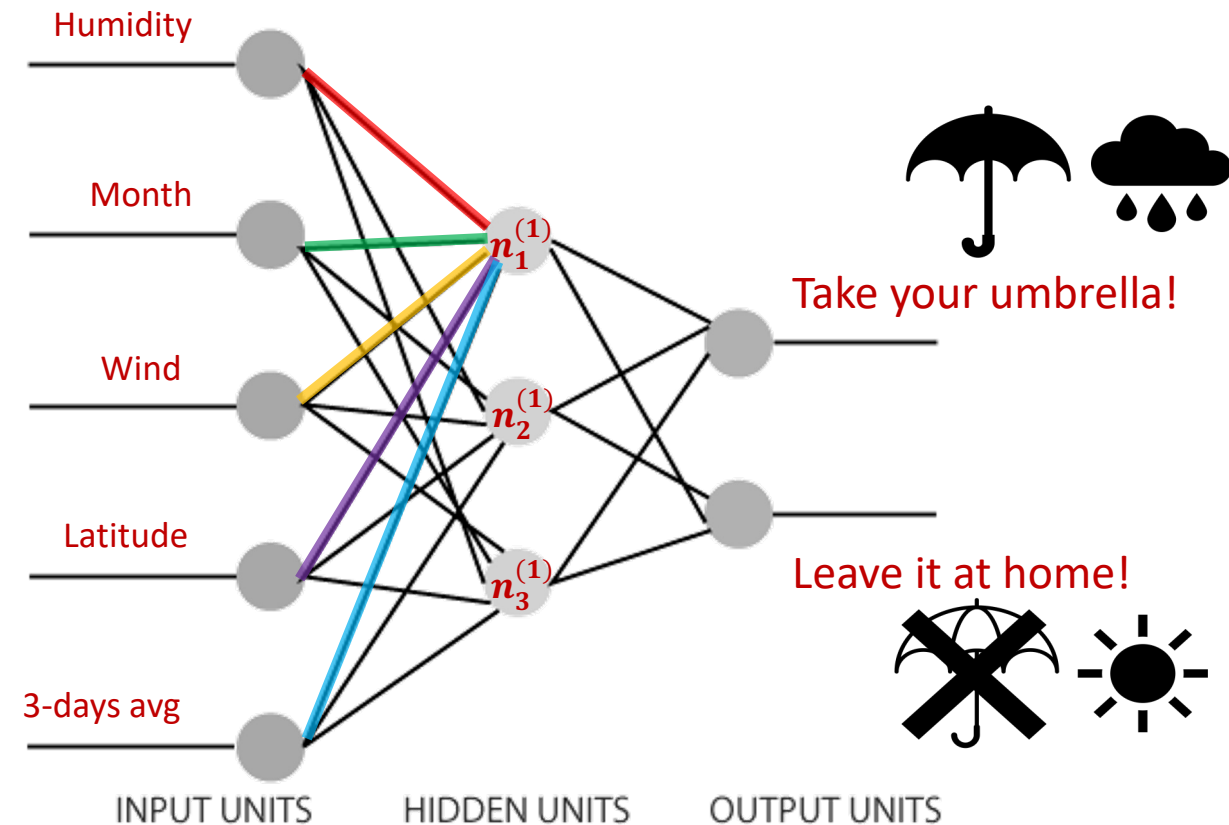
Example (weather forecast)

$$n_1^{(1)} =$$

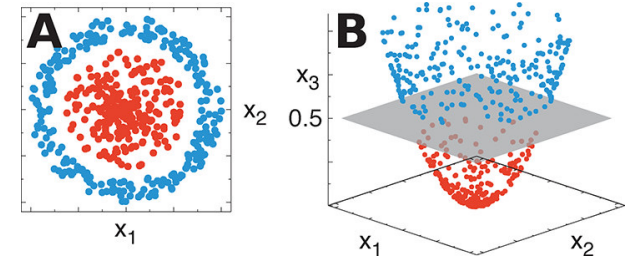
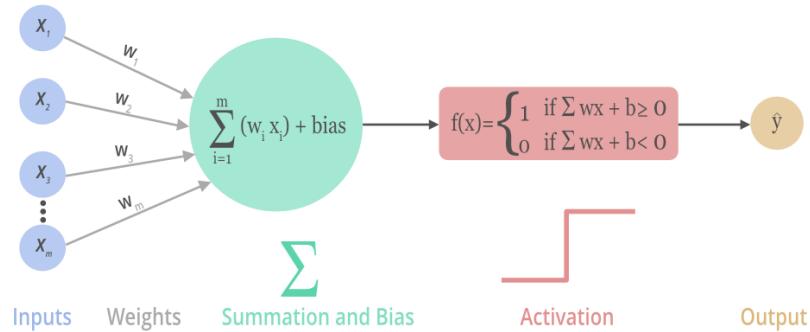
$$(\text{Humidity} \times W_{11}) + (\text{Month} \times W_{21}) + (\text{Wind} \times W_{31}) + (\text{Latitude} \times W_{41}) + (\text{3-days avg} \times W_{51})$$

$$W = [0.1, 0.15, 0.19, 0.005, 0.21]$$

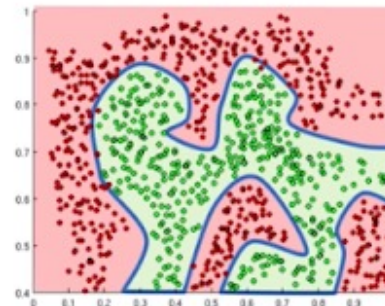
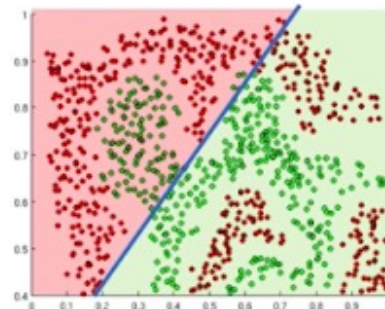
Do I need umbrella tomorrow?



Why Activation Function is needed?

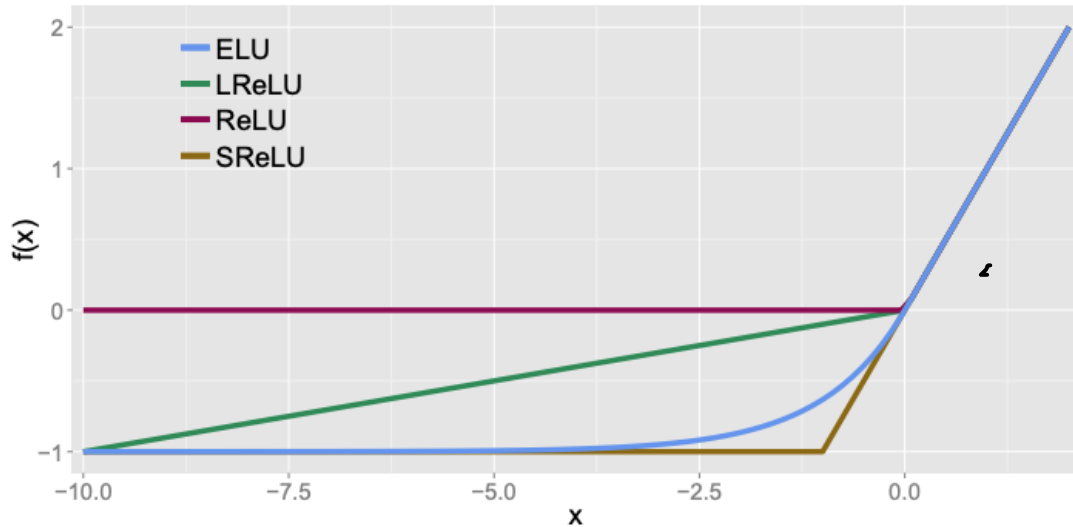


Simply put, an **activation function** is a **function** that is added into an artificial neural network in order to help the network learn complex patterns in the data.



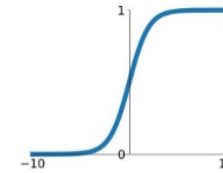
We can solve this problem using hidden layers with non-linear activation functions

Activation Function



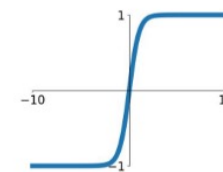
Sigmoid

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



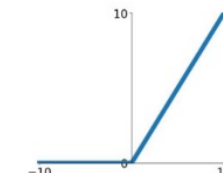
tanh

$$\tanh(x)$$



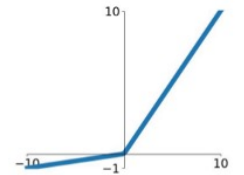
ReLU

$$\max(0, x)$$



Leaky ReLU

$$\max(0.1x, x)$$

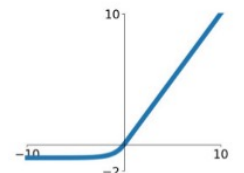


Maxout

$$\max(w_1^T x + b_1, w_2^T x + b_2)$$

ELU

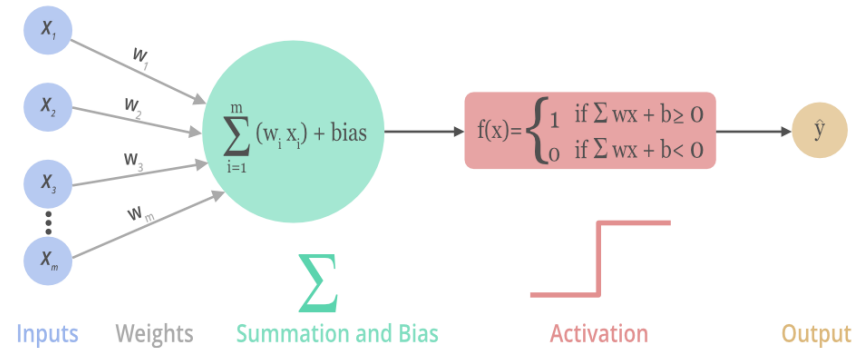
$$\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$$



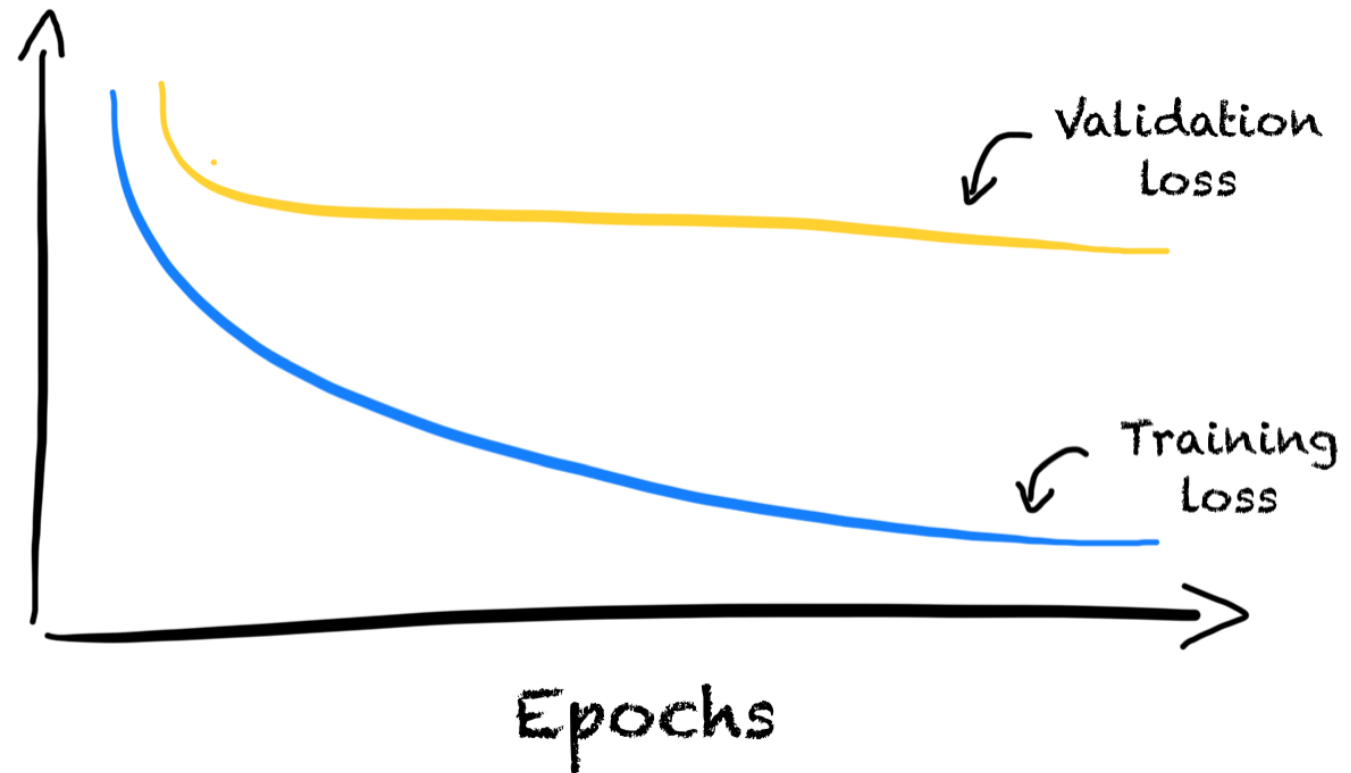
How do the model learn from data?

Just like human: we learn by being told what we're doing right or wrong, we all use feedback, all the time. Then, we change our approach next time and monitor the outcome.

A Neural Network has to figure out the **most optimum weights** for which the model is most reliable. Therefore, training means fining the most optimum values for weights (**W**).

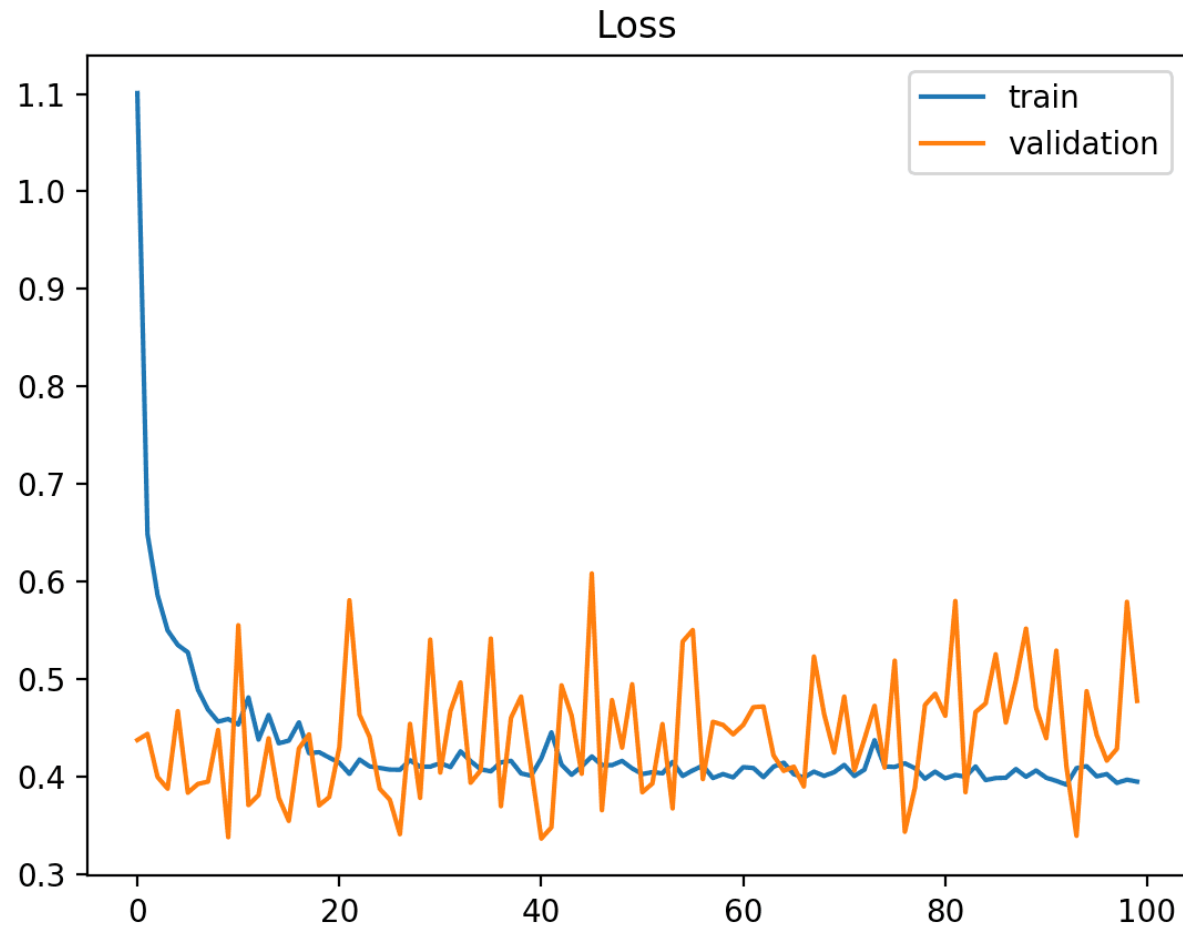


Learning Curve

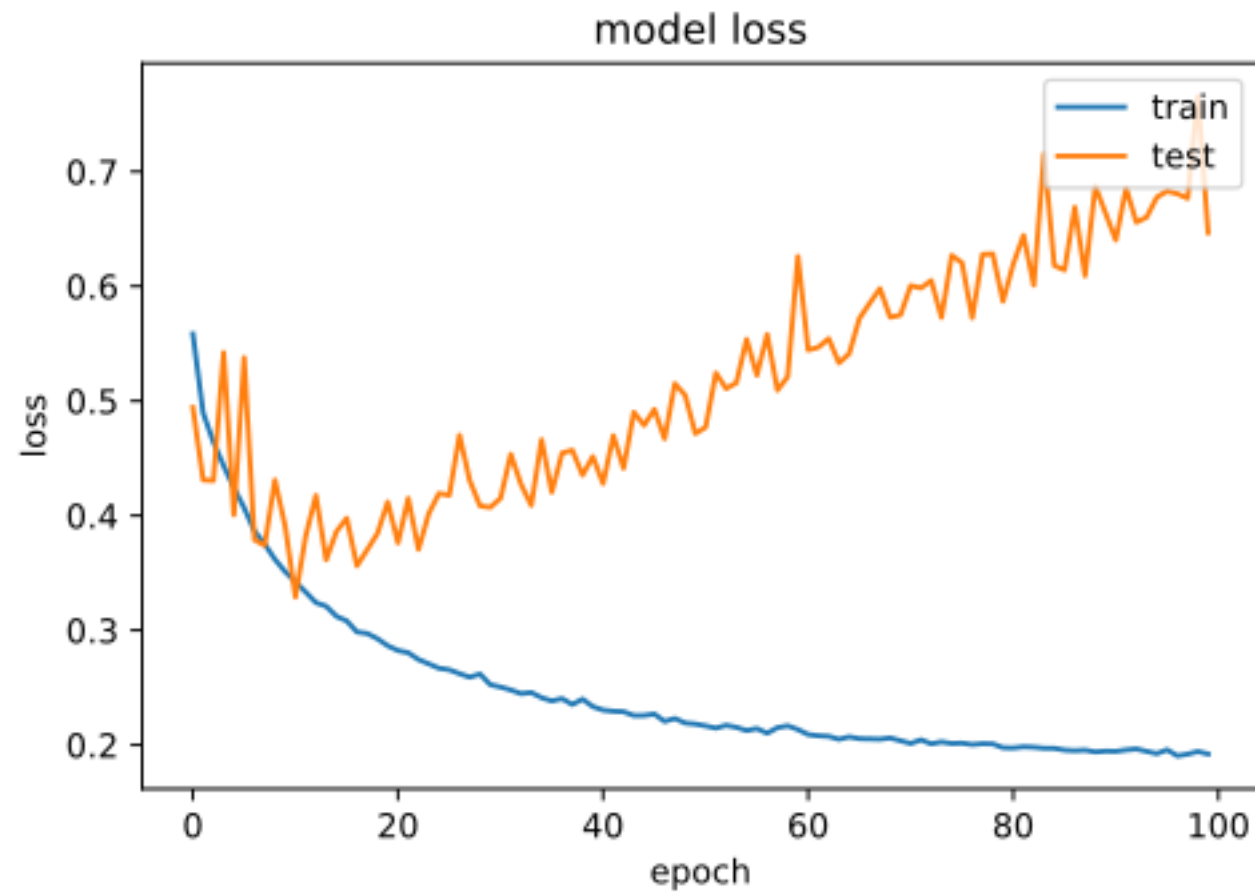


One the best ways to diagnose your model!

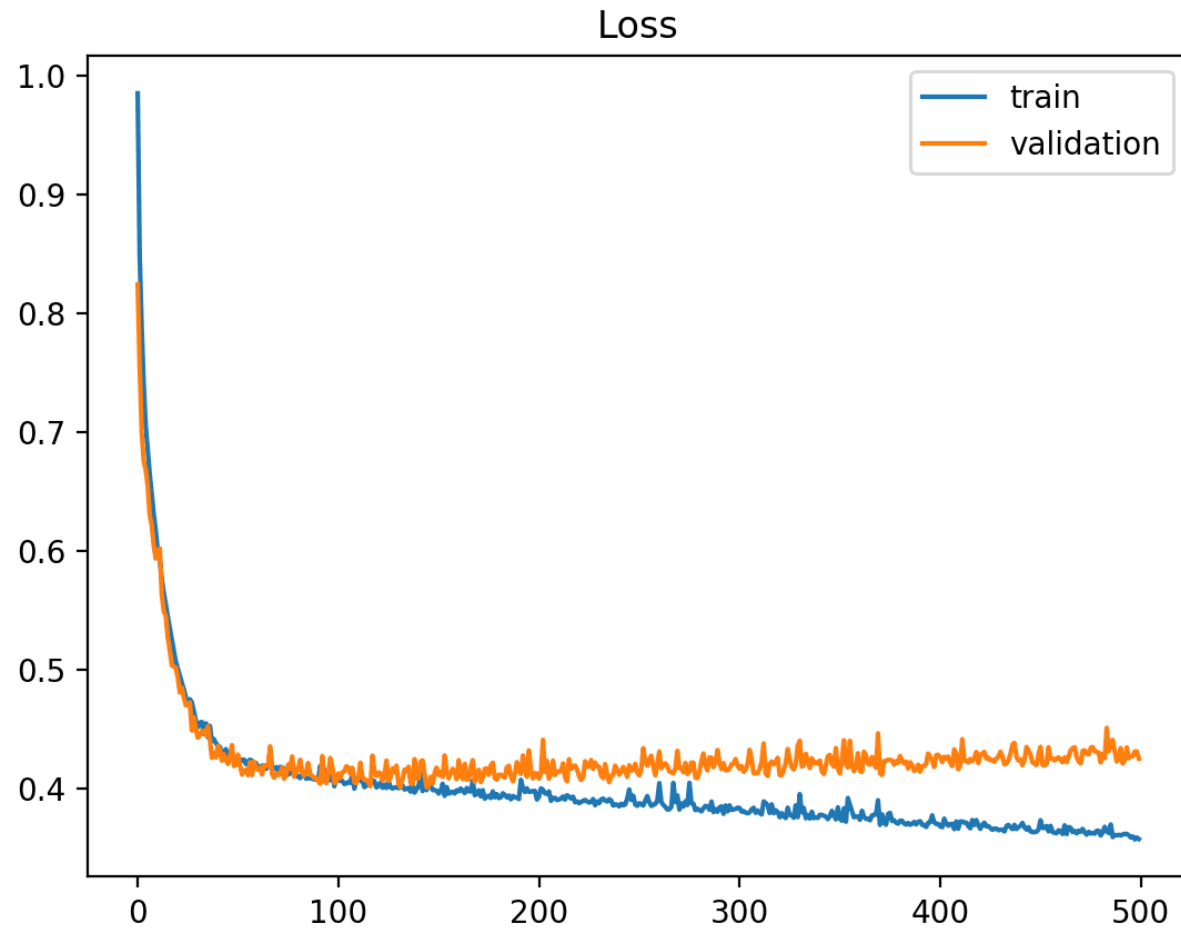
Learning Curve



Learning Curve



Learning Curve



Learning Curve

